STAFF & HR MANAGEMENT MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · HR Department · School Administrators

Related Modules:
Payroll · Finance · Attendance · Academics · Super Admin · Security & Access Control

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1. MODULE OVERVIEW

1.1 Purpose

The **Staff & HR Management Module** manages all employee-related operations within the institution.

This module handles:

- Staff records
- Recruitment
- Attendance tracking
- Leave management
- Performance evaluation
- Department management
- Staff analytics
- Contract lifecycle management

The system ensures **efficient human resource administration, compliance, and staff performance monitoring**.

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2. CORE OBJECTIVES

The module enables schools to:

1. Maintain **complete staff records**
2. Track **staff attendance**
3. Manage **leave and absence**
4. Monitor **staff performance**
5. Track **contracts and employment lifecycle**
6. Manage **departmental staff allocation**
7. Generate **HR reports and analytics**
8. Integrate **HR with payroll and finance systems**

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Super Admin	Platform-level HR management
School Admin	Manage staff records
HR Manager	Full HR operations
Department Head	View and manage department staff
Staff Member	View personal records
Auditor	Read-only access

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4. MODULE ARCHITECTURE

The Staff & HR Module consists of the following subsystems:

```
Staff & HR Management
│
├── Staff Dashboard
├── Staff Directory
├── Staff Profiles
├── Recruitment Management
├── Attendance Management
├── Leave Management
├── Performance Evaluation
├── Contract Management
├── Department Management
├── HR Reports & Analytics
├── HR Notifications
```

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5. STAFF DASHBOARD

5.1 Overview

The **Staff Dashboard** provides a real-time overview of the workforce.

Dashboard Metrics

- Total Staff
- Teaching Staff
- Non-Teaching Staff
- Active Staff
- Staff on Leave
- Staff Attendance Rate

Example Dashboard

```
-------------------------------------------
Staff Dashboard
-------------------------------------------

Total Staff: 120
Teaching Staff: 85
Non-Teaching Staff: 35

Active Staff: 112
On Leave: 8

Average Attendance: 94%
```

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6. STAFF DIRECTORY

Provides a searchable list of all staff members.

Features

- Staff search
- Department filtering
- Role filtering
- Contact directory

Example Directory Entry

```
Name: Jane Smith
Position: Mathematics Teacher
Department: Academics
Email: jane.smith@school.edu
Phone: +254 XXX XXX
Status: Active
```

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7. STAFF PROFILES

Each employee has a detailed profile containing:

- Personal information
- Employment details
- Contact information
- Qualifications
- Documents

Staff Profile Data Model

```
Staff
- staff_id
- full_name
- department_id
- role
- employment_type
- hire_date
- contract_end_date
- contact_information
- qualifications
```

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8. RECRUITMENT MANAGEMENT

Handles hiring processes.

Recruitment Workflow

```
Job Opening Created
        ↓
Applications Received
        ↓
Candidate Screening
        ↓
Interviews
        ↓
Hiring Decision
        ↓
Staff Record Created
```

Recruitment Data Model

```
JobApplication
- application_id
- candidate_name
- position
- resume
- application_status
```

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9. ATTENDANCE MANAGEMENT

Tracks staff attendance.

Attendance methods may include:

- Biometric scanners
- RFID cards
- Mobile check-in
- Manual entry

Attendance Data Model

```
StaffAttendance
- attendance_id
- staff_id
- check_in_time
- check_out_time
- attendance_status
- date
```

Attendance statuses:

- Present
- Absent
- Late
- On Leave

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10. LEAVE MANAGEMENT

Allows staff to apply for leave.

Leave Types

- Annual leave
- Sick leave
- Maternity leave
- Study leave
- Emergency leave

Leave Workflow

```
Staff submits leave request
        ↓
Department head review
        ↓
HR approval
        ↓
Leave recorded
```

Leave Data Model

```
LeaveRequest
- leave_id
- staff_id
- leave_type
- start_date
- end_date
- approval_status
```

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11. PERFORMANCE EVALUATION

Tracks staff performance metrics.

Evaluation Criteria

- Teaching effectiveness
- Student feedback
- Attendance
- Administrative contributions

Performance Data Model

```
PerformanceReview
- review_id
- staff_id
- evaluation_period
- rating
- comments
```

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12. CONTRACT MANAGEMENT

Tracks staff employment contracts.

Example:

```
Employee: John Doe
Position: Physics Teacher
Contract Start: Jan 2024
Contract End: Dec 2026
```

Contract Data Model

```
StaffContract
- contract_id
- staff_id
- contract_type
- start_date
- end_date
- salary
```

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13. DEPARTMENT MANAGEMENT

Organizes staff into departments.

Example departments:

- Academics
- Administration
- Finance
- Maintenance
- IT Support

Department Data Model

```
Department
- department_id
- department_name
- department_head
```

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14. HR REPORTS & ANALYTICS

HR Reports

- Staff attendance report
- Department staff distribution
- Staff performance report
- Leave utilization report

Example Report

```
Attendance Report (March)

Total Staff: 120
Average Attendance: 93%

Most Absences:
Maintenance Department
```

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15. STAFF NOTIFICATIONS

The system sends alerts for:

- Leave approvals
- Attendance alerts
- Contract expiry
- Performance reviews

Notification channels:

- In-app
- Email
- SMS

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16. SYSTEM INTEGRATIONS

Module	Integration Purpose
Finance	Salary management
Payroll	Salary processing
Attendance	Staff check-in system
Academics	Teacher assignments
Super Admin	Multi-school HR management

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17. SECURITY & ACCESS CONTROL

Security features include:

- Role-based permissions
- Encrypted staff data
- Secure document storage
- Multi-factor authentication

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18. AUDIT TRAIL

All HR actions are logged.

Example:

```
AuditLog
- log_id
- user_id
- action
- staff_id
- timestamp
```

Example entry:

```
User: HR Manager
Action: Updated Staff Profile
Staff: Jane Smith
Date: 16 March 2026
```

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19. SCALABILITY

The system supports:

- Hundreds or thousands of staff
- Multi-campus schools
- Cloud-based deployment
- High-volume attendance data

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20. MOBILE HR MANAGEMENT

Mobile support allows staff to:

- Check attendance
- Apply for leave
- View schedules
- Receive notifications

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21. BULK DATA MANAGEMENT

Supports:

- Staff data import via CSV/Excel
- Bulk updates
- Report exports

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22. FUTURE ENHANCEMENTS

Advanced capabilities:

- AI workforce analytics
- Staff burnout prediction
- Smart attendance anomaly detection
- AI recruitment screening
- Automated performance evaluation